

L'écosystème mangrove, services écosystémiques, protection des zones côtières et restauration.



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Comment restaurer un écosystème précieux et menacé ?

- I. Qu'est-ce que la mangrove ?
- II. Quels services écosystémiques rend-elle ?
- III. Quelles pressions la menacent ?
- IV. Comment la restaurer ?

An aerial photograph showing a dense, green mangrove forest in the foreground, transitioning into a shallow, turquoise ocean. The water's surface is textured with ripples and small waves. A semi-transparent dark rectangle is overlaid on the image, containing the title and subtitle text.

I Qu'est-ce que la mangrove ?

écosystème - distribution - évolution

Mangrove : Forêt (sub)tropicale de la zone intertidale

Biotope : deltas, côtes abritées

- Substrat : vasières littorales
- Hydrologie : macrotidale ou estuarienne

Palétuviers : 70 sp. adaptées au milieu

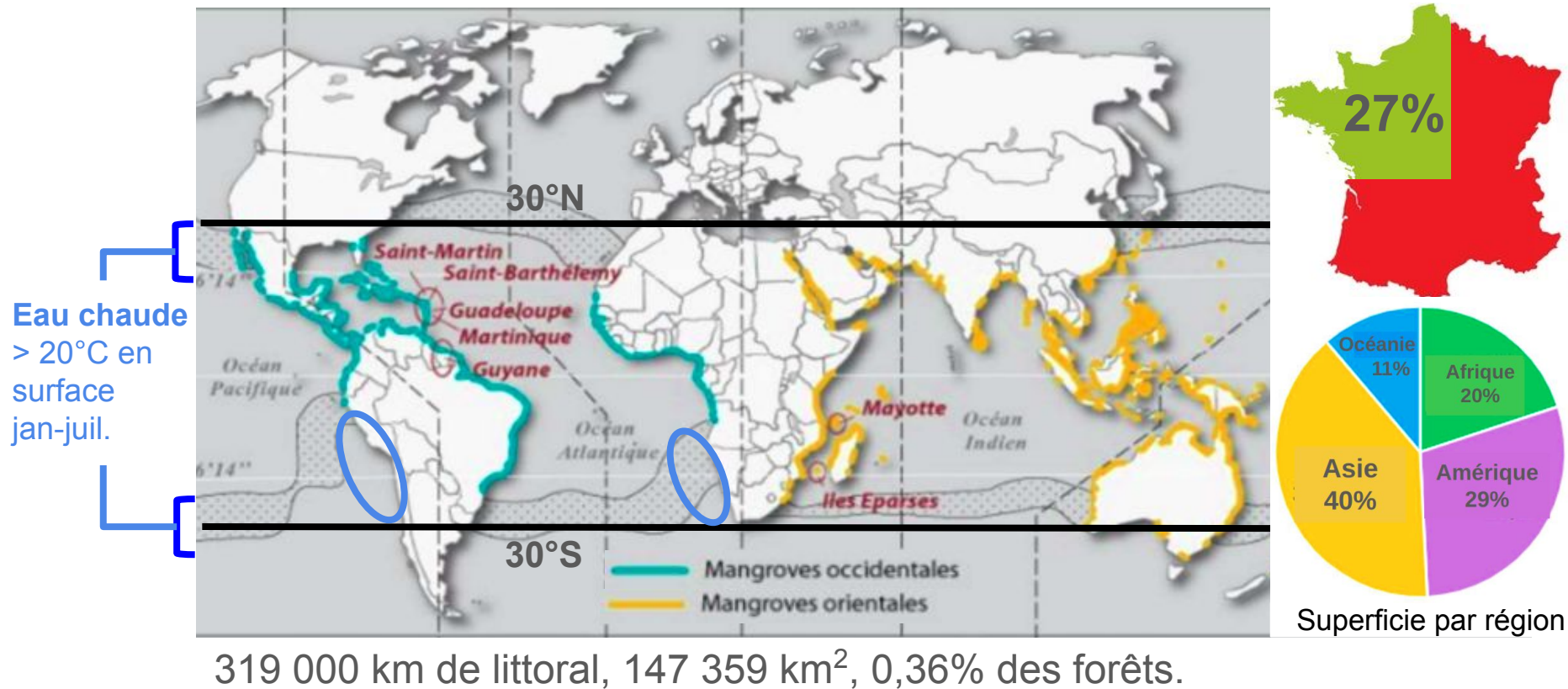
- Racines aériennes, Propagules
- Zonation par hydropériode et salinité

Socio-écosystème

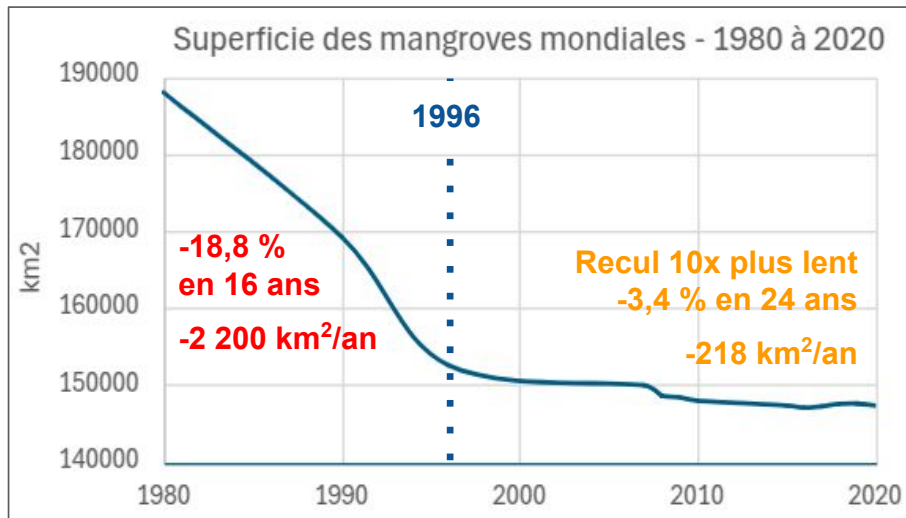
- Biocénose très riche
- Communautés locales qui en dépendent



Distribution : 75% des côtes tropicales et subtropicales



Evolution : recul qui ralentit mais qui persiste

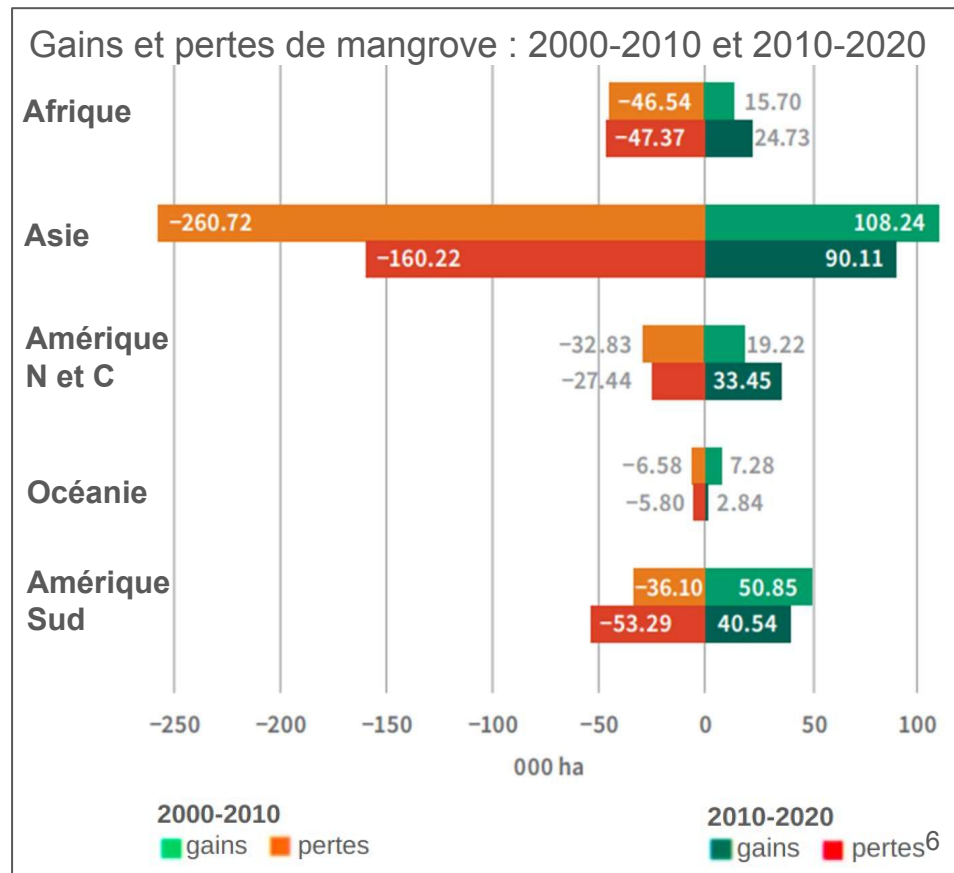


2000-2020 : gains > pertes

Pertes : ralentissement

Gains : stagnation

Sources : Bunting, *et al.*, 2022 ; FAO 2023.



The image is a composite of two photographs. The left side shows a dense mangrove forest with trees growing in shallow, rippling water. The right side shows a woman standing in a mangrove swamp, wearing a blue tunic and a colorful headwrap, holding a large bowl filled with harvested oysters. She is wearing gloves and smiling at the camera.

II Services écosystémiques

Protection - Régulation - Ressources

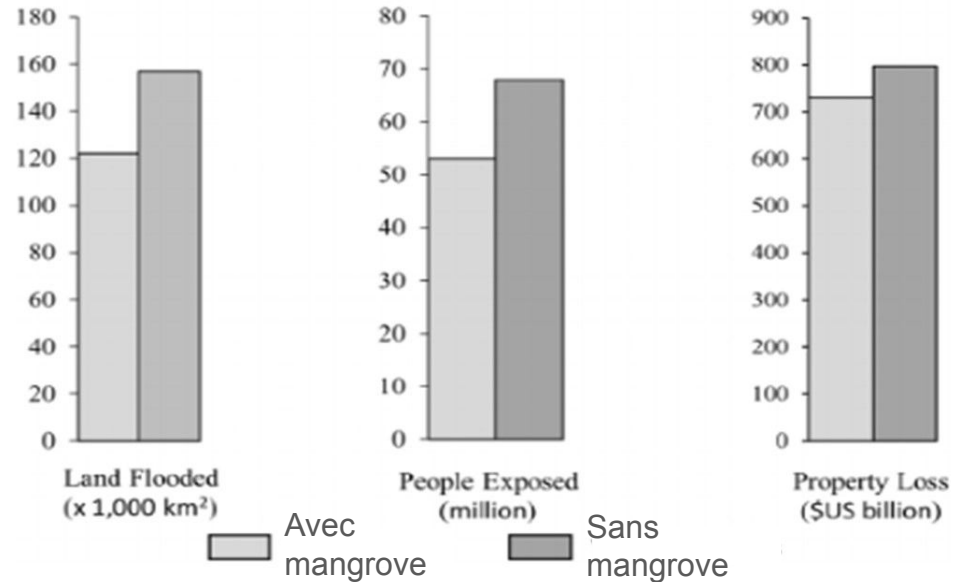
Protection des zones côtières

Limite les inondations (ouragans) :

- 15 M personnes / an
- 65 \$USB de biens / an
- 35 000 km² de terre / an

Protège le trait de côte :

- Eustatisme : Retient les sédiments
- Erosion : Amortit les vagues



Valeur annuelle attendue de la protection des mangroves contre les inondations. Menendez *et al.* 2020

Régule le climat et améliore la qualité de l'eau

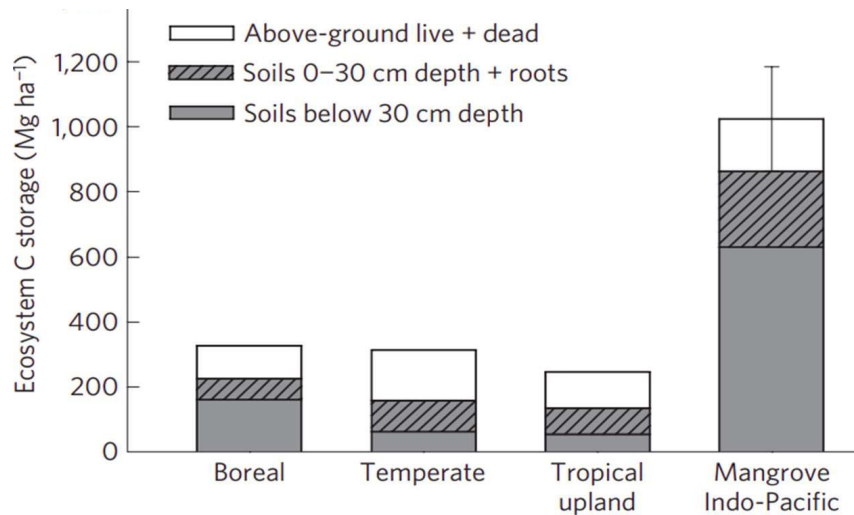
La forêt la plus dense en carbone

- Produit beaucoup de racines
- Décomposition lente : substrat anoxique
- Enfoui sous les sédiments

Filtre l'eau : arrête polluants et sédiments

- Protège les récifs
- Améliore la qualité de l'eau

Ex : Filtre l'azote : 109 M€ / an DOM TOM



Comparaison du stockage de carbone dans les mangroves avec celui des principaux domaines forestiers mondiaux (Donato *et al.*, 2011)

Ressources pour les écosystèmes et les humains

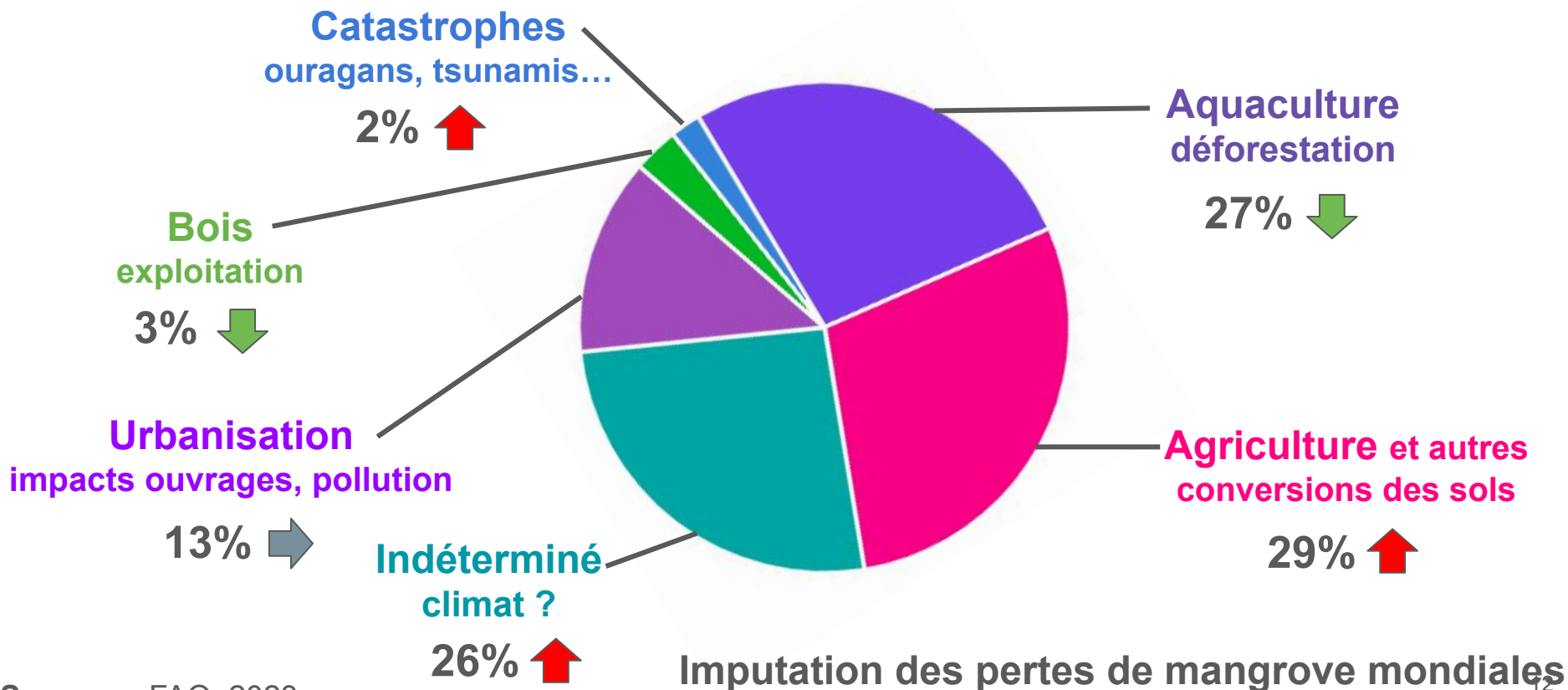
Biodiversité	Nurserie et abri journalier : herbiers et récifs
Pêche côtière	Mexique : 1 ha déforesté = - 37 500 \$ / an pour les pêcheries
Aquaculture	Indonésie : +18% crevettes / ha si mangrove à moins de 100m
Alimentation	Indonésie : Habitants des mangroves : +28% poisson et fr. de mer
Valorisation	Bois, éco-tourisme



III Menaces

Pressions anthropiques - changement climatique

Pressions anthropiques et environnementales



Impacts présents et à venir du changement climatique

3 effets négatifs	
Eustatisme	Mangroves des petites îles et lagons <i>Pacifique, caraïbes</i>
Canicules et Sécheresses	Mangroves des delta et des côtes arides <i>Afrique, Asie du sud, Mexique, Australie</i>
Ouragans	Toutes les mangroves
1 effet positif	
T° plus chaudes	La mangrove s'étend vers les hautes latitudes <i>Chine, USA, Australie, Nouvelle Zélande</i>

A man with short dark hair, wearing a white t-shirt and a tan field vest, is standing in a mangrove wetland. He is looking down at a small mangrove plant he is holding in his hands. The vest has two patches: one on the left chest with a blue logo and the word 'KINSHASA' below it, and another on the right chest with a green logo and the word 'KINSHASA' below it. He is also wearing a black watch on his left wrist. The background shows a vast expanse of mangrove trees with their roots visible in the water under a clear sky.

IV Restauration

Etat de l'art, retour du terrain et pistes d'amélioration

Etat de l'art : pourquoi de nombreux projets échouent ?

Choisir un site propice

Site de mangrove détruite ou dégradée



30% de Vasière nue : survie 1,4%
Ecosystème installé : herbier

Diagnostic socio-écologique, comparatif avec un site intact

Indispensable avant toute restauration.



Réalisé dans 6% des projets

Réduire les pressions

Créer des zones protégées
Habitants : revenus durables



87% projets : replantation seule

Réhabilitation ?

Rétablir l'hydrologie
Avant toute plantation



9% des projets réhabilitent l'hydrologie

Replantation ? Utile : 16% des cas

Espèces diversifiées, locales
Protéger les plantules



Monospécifique (*Rhizophora* sp.)
Taux de survie : 20%

Suivi, évaluation

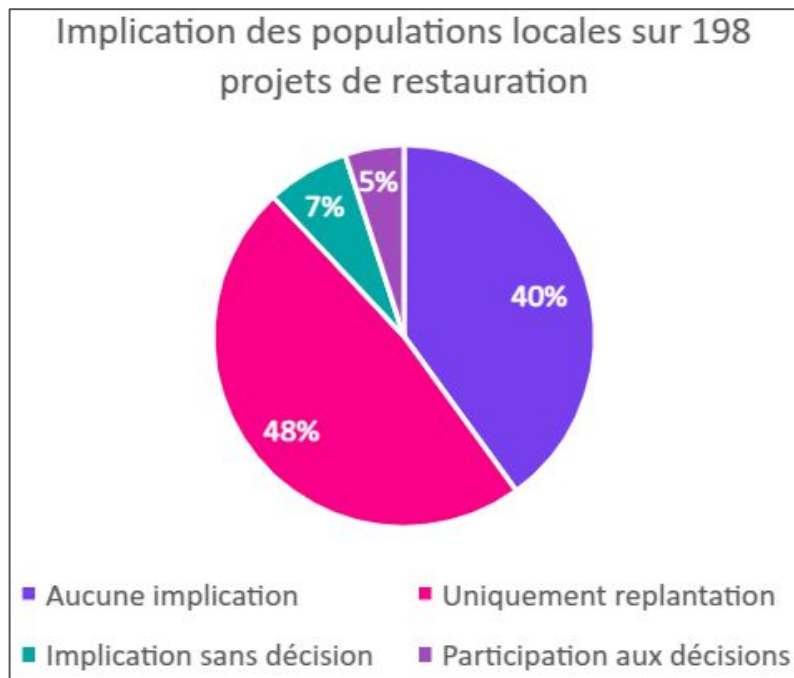
Durée : 5 ans



Majorité des cas : aucun suivi

Comment mieux gérer ces projets ?

Impliquer les populations locales



Utiliser des méthodes plus scientifiques

- Méthodes et indicateurs standardisés
- Publication des bilans de projets

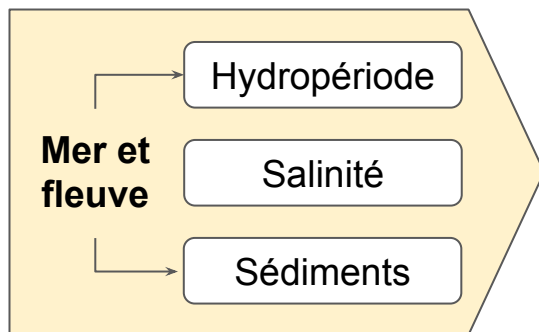
Financer en priorité :

- Les projets qui suivent l'état de l'art
- Les petits projets : plus efficaces

SERVICES ECOSYSTEMIQUES



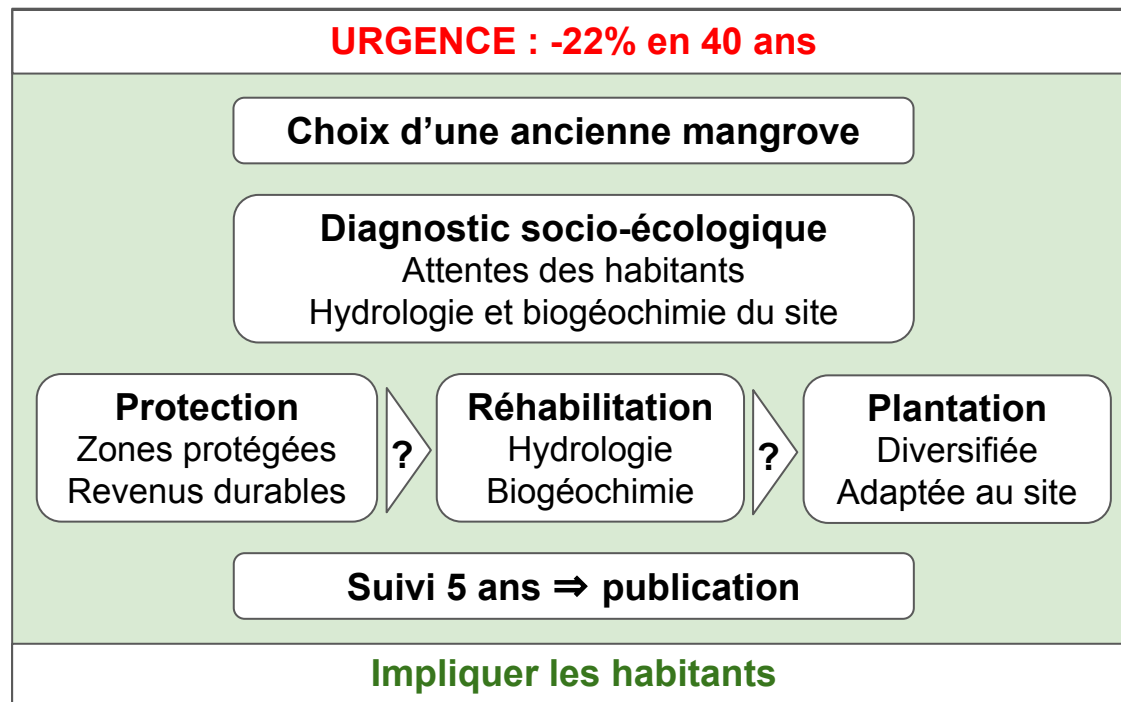
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An aerial photograph of a vast, dense green forest. A winding river flows through the center of the forest, reflecting the surrounding trees. A small boat is visible on the river in the lower-left quadrant. The word "Abstract" is overlaid in white text on a semi-transparent dark rectangle in the center of the image.

Abstract

Tropical and subtropical coastal forest

Cover 75% of tropical / subtropical coastline

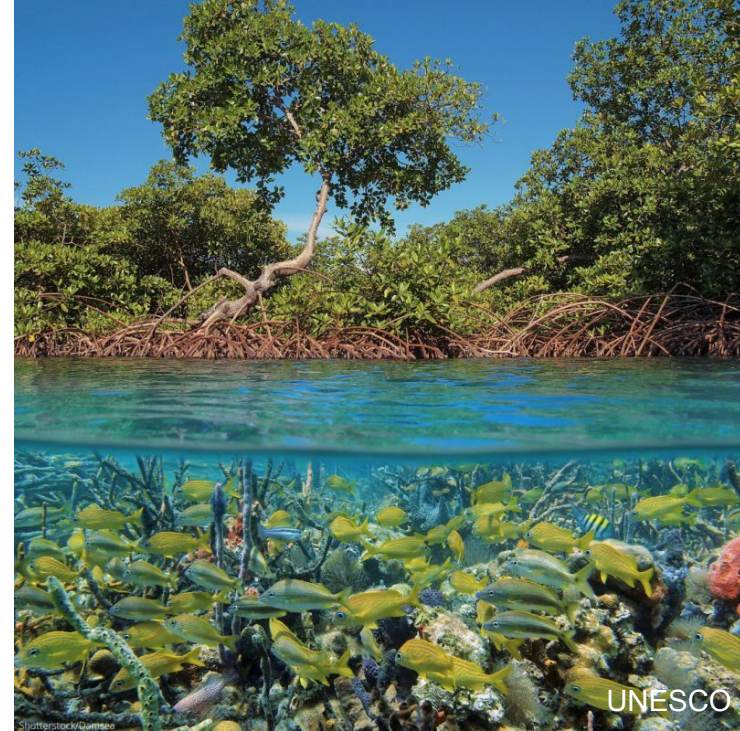
Have a small footprint : only 27% of France

Need specific conditions :

- Muddy sediment
- Roots flooding
- Low wave exposure

High levels of biodiversity

Eg: Fish, birds, reptiles, crabs, shrimps...



Mangroves provide valuable ecosystemic services

Protect coastlines from flooding and erosion

Store carbon very effectively

Filter pollution and sediments in sea water

Boost fishing and aquaculture yields

Provide food and revenue to local communities

Act as nursery and shelter for wildlife



FAO/Sylvain Cherkauoi

Mangrove are slowly declining

Deforestation for human activities :

- Aquaculture
- Agriculture
- Urbanisation

Environmental pressure :

- Climate change
- Dams and irrigation

Restoration results do not compensate



How can we restore mangroves better ?

Most projects do not follow best practices :

- 94% do no initial assessment
- 87% **only plant trees**
- 91% **don't fix hydrology**
- 95% **decide without local communities**
- Most **don't monitor results**

So, there is much to improve :

- **Use more science**
- **Follow best practices**
- **Monitor and publish results**



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